Ahmed Aissa Berraies

Work permit: Belgian, Italian, Dutch

Date of birth: 01/01/1992

Place of birth: Ras Jebel, Tunisia

Nationality: Tunisian

Gender: Male

Contact

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europass

ABOUT MYSELF

PhD-level computational mechanics consultant with a proven track record of bridging advanced numerical research and practical industrial Multiphysics applications. Specializing in structural mechanics, computational fluid dynamics (CFD), fluid-structure interaction (FSI). Combines deep technical proficiency in simulation tools with data science & AI skills using best management practices to lead complex engineering projects and optimize product designs.

WORK EXPERIENCE

Independent consultant Brussels, Belgium

Lead Multiphysics simulation Consultant

01/01/2026 - Current

- Accelerate R&D and product optimization by integrating High-Performance Computing (HPC) with advanced multi-physics simulations, including structural, CFD and FSI. Strong foundation in machine learning, data science and AI.
- Drive data-informed engineering solutions by deploying custom workflows and techniques to transform massive datasets generated from complex simulations and physical tests into actionable, commercial insights.
- Prioritize tasks, manage project scope, and align cross-functional stakeholder expectations to ensure seamless integration between advanced computational research and practical business objectives.

Université Catholique de Louvain Louvain-la-Neuve, Belgium

Simulation expert

01/01/2025 - 15/07/2026

- Engineered a novel 3D-structured porous transport electrode (PTE) concept to optimize efficiency and performance in alkaline water electrolysis systems.
- Developed end-to-end computational workflows, translating X-ray micro-computed tomography images of foam electrodes into advanced CFD simulations to drive cell design iterations.
- Leveraged data science methodologies and Python scripting to systematically process X-ray tomography outputs, design, test, and validate the performance of new porous structures, extracting actionable engineering insights from complex datasets.

University of Pavia PAVIA, Italy

Engineering research scientist

01/04/2022 - 31/03/2024

- Executed advanced computational fluid-structure interaction (FSI) simulations to conduct rigorous functional assessments of novel vascular prostheses.
- Evaluated the performance and viability of innovative vascular and orthopedic implants produced via cutting-edge additive manufacturing (3D printing) techniques.

ProTechTion Marie-Sklodowska Curie: University of Pavia & Eindhoven University of Technology Pavia, Italy

Early-Stage Researcher

01/12/2018 - 31/01/2022

- Partnered with industrial manufacturer to deliver a robust design tool for microfluidic valve actuation control.
- Solved complex product development bottlenecks by deploying a full suite of computational Multiphysics methods, seamlessly integrating CFD, FSI, and nonlinear solid mechanics.
- Enhanced simulation accuracy and product reliability for commercial application through the advanced calibration of complex hyperelastic materials.

Engineering Procurement and Project Management (EPPM) Tunis, Tunisia

Mechanical engineer/ Piping and pipeline

01/10/2017 - 31/10/2018

- Oil & Gas piping engineering & piping modeling and support design analysis using CAEsar II.
- Water and sewage treatment plants engineering (ARAMCO).
- Sea Water desalination technical evaluation and commissioning.
- Design of storage tanks according to API650 for a electric power plant.

Université Catholique de Louvain, Belgium Louvain-la-Neuve, Belgium

Master of Engineering intern

23/03/2017 - 04/09/2017

- Micromechanical modeling and simulation of composite materials using FEA.
- Prediction of the macroscopic behavior of a composite materials using multi-scale methods.

EDUCATION & TRAINING

01/12/2018 - 03/09/2025 Eindhoven, Netherlands

Doctor of philosophy, PhD Universita di Pavia, Italy/ Eindhoven University of Technology, the Netherlands

- Partitioned fluid-structure-contact interaction
- Computational fluid dynamics
- Nonlinear solid mechanics
- Reverse engineering
- Design and modeling

Website: https://www.tue.nl/en/news-and-events/news-overview/03-09-2025-developing-a-computational-model-of-a-pilot-operated-control-valve-with-advanced-numerical-methods**Field(s) of study:** Computational mechanics | **Final grade:** Joint PhD diploma | **Thesis:** Numerical methods for fluid-structure interaction in a pilot-operated valve

03/09/2013 - 30/09/2017 Tunisia

Mechanical engineering National Engineering School of Monastir (ENIM)

Address: Monastir (Tunisia) | **Website:** https://enim.rnu.tn/ **Field(s) of study:** Mechanical Engineering

03/09/2010 - 30/05/2013

Bachelor of engineering (Mathematics and physics) Institut préparatoire aux études d'ingénieurs de Tunis (IPEIT)

Address: Tunis **Field(s) of study:** Mathematics & Physics

SKILLS

Microsoft Office | Agile project management | Project planning and management | Strong international business experience; collaboration with cross-cultural customers & colleagues | Python | MATLAB | Computational Fluid Dynamics (CFD) | Numerical methods | Computer Aided Design | LaTeX | Linux | High performance computing (HPC) | ANSYS | project management | create solutions to problems | think analytically | think creatively | make numerical calculations | prioritise tasks | work as a team | data science | statistics | machine learning | Artificial İntelligence

LANGUAGE SKILLS

MOTHER TONGUE(S): Arabic

OTHER LANGUAGE(S):

French

Listening C2
Reading C1
Writing C1
Spoken production C1
Spoken interaction C1

English

Listening C1
Reading C1
Writing B2
Spoken production B2
Spoken interaction B2

Italian

Listening B1
Reading B1
Writing A2
Spoken production B1
Spoken interaction B1

CONFERENCES & SEMINARS

27/05/2019 - 29/05/2019 El Campello, Alicante, Spain
International Conference on Adaptive Modeling and Simulation, ADMOS 2019
Link congress.cimne.com/admos2019

11/01/2021 - 15/01/2021 Paris, France
WCCM-ECCOMAS CONGRESS 2020
Link wccm-eccomas2020.org

01/09/2021 - 03/09/2021 Pavia, Italy
Sim-AM2021
Link congress.cimne.com/sim-am2019

21/05/2019 - 23/05/2019 Milano, Italy
HPC for Industry 4.0
Link events.prace-ri.eu

2020 Online
Ansys Simulation World

PUBLICATIONS

2025
Numerical analysis of single-phase electrolyte transport properties in 3D porous electrode structures (Submitted)
Authors: Ahmed Aissa Berraies | **Journal Name:** Chemical Engineering Science | **Publisher:** Elsevier

2025
On the Robustness of Dirichlet–Neumann Coupling Schemes for Fluid-Structure-Interaction ProblemsWith Nearly-Closed Fluid Domains
International Journal for Numerical Methods in Engineering
Authors: Ahmed Aissa Berraies | **Journal Name:** International Journal for Numerical Methods in Engineering

Volume, Issue and Pages: Volume126, Issue18 | **Publisher:** Wiley

2024

Numerical investigation of fluid–structure interaction in a pilot-operated microfluidic valve

Authors: Ahmed Aissa Berraies | **Journal Name:** Journal of Fluids and Structures | **Publisher:** Elsevier

2019

Short fiber reinforced composites: Unbiased full-field evaluation of various homogenization methodsin elasticity (Co-author)

Journal Name: Composites Science and Technology | **Publisher:** Elsevier

MANAGEMENT AND LEADERSHIP SKILLS

29/09/2025 - Current

PMI Project Management Professional (PMI-PMP)

26/02/2024 - 01/03/2024

PMI Professional in Business Analysis (PMI-PBA)

PSM I Certified (Professional Scrum Master I)

PSPO I Certified (Professional Scrum Product Owner I)